

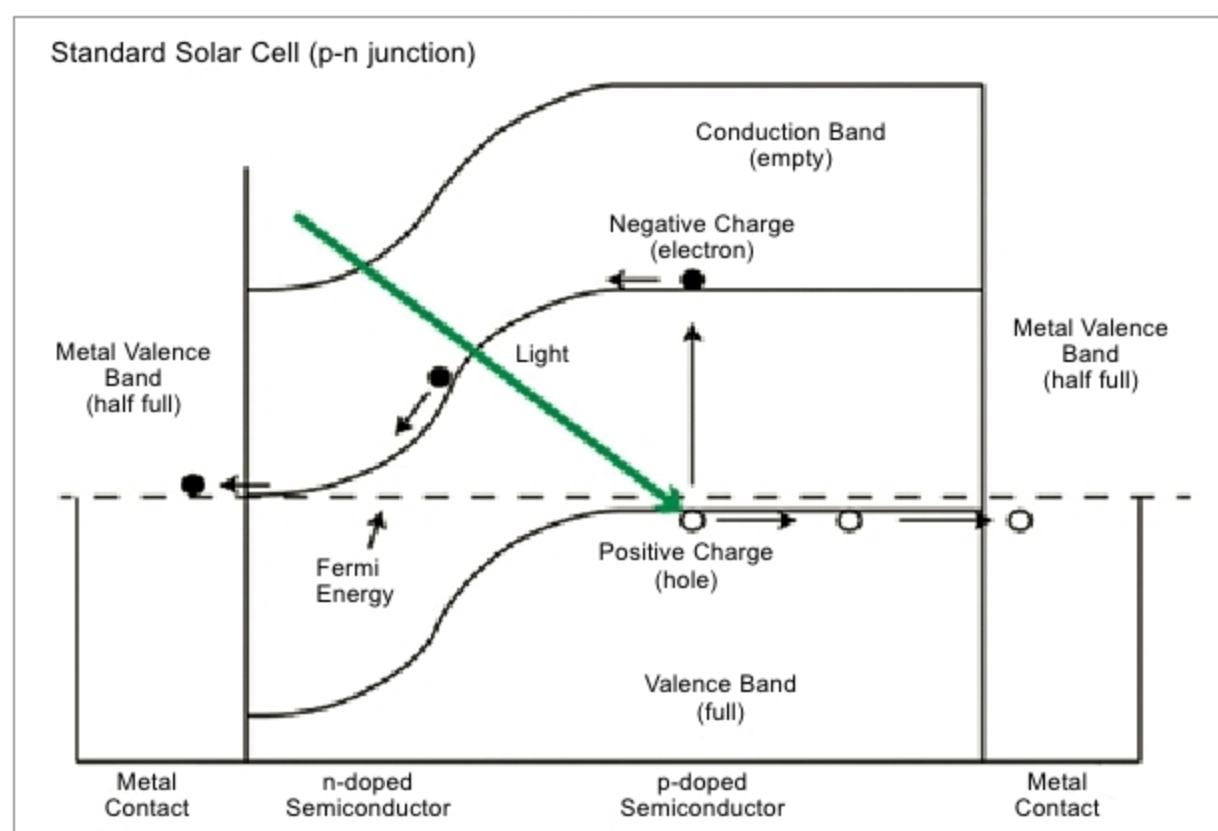


Fig. 2: Mechanism of generation of electrons

Band Gap Values of Some Selected Materials Table 1

Group	Material	Band gap (eV) @302K
IV	Silicon	1.14
IV	Germanium	0.67
III-V	Gallium nitride	3.4
III-V	Gallium phosphide	2.26
III-V	Gallium arsenide	1.43

electrical potential difference of one volt in a vacuum. Band gap values of some selected materials are shown in Table 1.

Loss mechanism in solar cells

Multi-junction solar cells with concentrator can have over seventy per cent theoretical efficiency. But various mechanisms given below result in a chasm between efficiencies promised from theoretical calculations and those reached in practice.

Reflection loss. The reflection loss occurs from the top surface of the cells. Photons striking the top surface are reflected due to high reflectivity in the UV and visible regions, resulting in the absorption of a small portion of incident light leading to poor efficiency. These losses account for about thirty per cent. To reduce the reflection loss, a common approach is the use of an antireflective coating (ARC) of about 60nm thickness in the form of inverted pyramids by etching

the surface with acid.

Recombination loss. Photon incident on the solar cells generates electron-hole pairs called carriers. These carriers need to be separated before they recombine with the emission of energy. Recombination causes loss of carriers and affects the performance of the cell.

Series resistance loss. Power losses in the solar cell take place due to a combination of resistance in series. Series resistance losses contribute to around twenty per cent of the total input power. But these losses increase tremendously when solar cells are operated at high intensities using lenses or mirrors.

Heavily doping of the semiconductor region helps in generating surface fields, which reduce the recombination at the metal-semiconductor contact. This is called tunnelling because the carriers can just tunnel through the junction. Other measures like the use of high conductivity substrate material, optimising the junction depth, increasing the number of fingers, using silver or gold, or electroplating metal contacts with these precious metals reduce the resistance.

Thermal loss. A major portion of the loss in a solar cell is due to heat. Light absorbed by solar cells has excess energy than that required for the generation of carriers. This excess energy is released in the form of heat. Proper cooling arrangements are used to main-

tain the temperature within limits.

Spectral mismatch loss. Single-junction solar cells suffer from intrinsic efficiency losses due to spectral mismatch of various wavelengths of light. The inability to absorb photons with an energy lower than the band gap and relaxation losses of the excess energy of absorbed high energy photons make up about 65 per cent of the total energy loss in a single-junction solar cell. The use of multiple p-n junctions made of different semiconducting materials with matching band gap energy allows the absorbance of a broader range of wavelengths, thereby improving the cell's sunlight to electrical energy conversion efficiency.

Multi-junction (MJ) solar cells

Multi-junction or stacked solar cells are currently the most efficient cells available that convert up to 45 per cent of the solar energy they absorb into electricity under concentrated sunlight. The high efficiency of these cells stems from their ability to utilise a broad range of the solar spectrum from 300nm to beyond 1800nm, while controlling the spectrum incident on each subcell to minimise the energy loss of photogenerated carriers as these thermalise to the band's edges.

MJ solar cells are primarily constructed of III-V semiconductor materials. These semiconductor materials are from groups III and V of the periodic table of chemical elements.

Theoretically, an infinite number of junctions would have a limiting efficiency of 86.8 per cent under highly concentrated light, implying that more junctions increase efficiency. The maximum theoretical efficiency is 37, 50, 56, and 72 per cent for 1, 2, 3, and 36 p-n junctions, respectively.

As the cell approaches the limit of efficiency, the increase in cost and complexity grows rapidly. MJ solar cells are solar cells with multiple p-n junctions made by stacking different semiconductors with varying band gaps on top of one another.

Each material's p-n junction will produce an electric current in response to different wavelengths of light, improving the cell's sunlight to electric energy conversion efficiency. However, these cells are ten times more costly than less efficient thin-film silicon